

1. Data Construction

Antioxidant Food Table

Carlsen et al. 2010 (CC-BY)
138-page PDF supplement



Structured Extraction

word bounding boxes
+ statistical validation



Modelling Table

3,135 records · log(1+FRAP)
product text + category

2. Leakage-Aware Split Design

Name Normalisation

39.4% of records share
a product name



Data Partitioning

5 seeds per regime
(fit / validation / 20% test)



Grouped by Name

no shared name
across partitions



Random Split

literature default
(37% test duplicates)

3. Model Development

Baselines

global median
category median
memorization
(duplicate lookup)

Deep Learning

char-CNN · BiLSTM
+ category embedding
→ 192-d representations



ML-DL Integration

embeddings + TF-IDF →
XGBoost / Ridge hybrids
fusion of 8 models



Classical ML

TF-IDF + category
Ridge · k-NN · RF
XGBoost · LightGBM · MLP

4. Evaluation and Diagnostics

Leakage-Aware Scoring

R^2 · RMSE · MAE · Spearman ρ on log(1+FRAP)
5 paired seeds · y-permutation controls



Diagnostics

$\Delta R^2 = \text{random} - \text{grouped} \cdot \text{memorization}$
decomposition · text / category ablations